

Nwoyr Alshahrani | Invoice Audit Evaluation

Executive Summary & Scope: I focused my development and evaluation on Hospital 1 (H1) and Hospital 4 (H4). For H4, I selected 227 predictions from 840 total records (7 flagged, 220 unflagged), omitting 613 records where complete pricing remained unresolved. Five selected totals differed from billed amounts. H2, H3, and H5 were placed outside my final execution scope.

H1 Performance Measurement

I evaluated 908 H1 invoices with unambiguous identifiers; 10 records sharing 5 invoice IDs were excluded. **Results:** TP: 35, FP: 0, FN: 18, TN: 855 | **Precision:** 100.00%, **Recall:** 66.04%, **F1:** 79.55%, **Accuracy:** 98.02%. Note that H1 informed my development process, so these metrics represent development validation rather than held-out H4 accuracy. A negative finding indicates no anomaly detected by my rules, rather than complete audit clearance.

H1 Category (Literal Name)	TP	FP	FN	Precision	Recall
bundle_not_applied	0	0	5	-	0.00
contract_number_mismatch	4	0	0	1.00	1.00
cross_invoice_duplicate	0	0	4	-	0.00
daily_cap_exceeded	0	0	4	-	0.00
exclusion_window_violation	0	0	3	-	0.00
invoice_total_mismatch	6	0	0	1.00	1.00
line_total_arithmetic	6	0	0	1.00	1.00
malformed_service_date	6	0	0	1.00	1.00
premium_incorrectly_applied	0	0	6	-	0.00
premium_omitted	0	0	3	-	0.00
service_date_after_invoice_date	5	2	0	0.71	1.00
service_date_out_of_window	5	0	0	1.00	1.00
unit_price_mismatch	3	0	7	1.00	0.30
unknown_service	0	0	12	-	0.00
volume_discount_incorrectly_applied	0	0	4	-	0.00
volume_discount_omitted	0	0	4	-	0.00
weekend_rate_mismatch	0	2	0	0.00	-
wrong_unit_basis	9	0	1	1.00	0.90

Systematic Limitations & Verification

Four Identified Limitations in My System:

- Identification:* Unresolved descriptions are deferred rather than declared unknown automatically (e.g., INV-H1-000667).
- Pricing Coverage:* Service matching does not guarantee adjusted price accuracy (e.g., INV-H1-000151 missed unit price mismatch).
- Daily Aggregation:* H1 daily-cap validation is not implemented, though H4 has a dedicated implementation.
- Cross-Record Logic:* H1 bundles, exclusions, and duplicates remain unimplemented.

Separate Calculation Check: I built a parallel execution check that recomputed all 227 selected H4 invoices (2,347 lines) directly from contract rules and raw JSONL. It yielded **0 disagreements** in totals and flags, confirming internal pipeline consistency.

Decision Log & Technical Architecture

Scope & Money Handling: I enforced deterministic, exact integer cent calculations using Decimal with half-up rounding. Any selected amount discrepancy is explicitly flagged as *recomputed_total_mismatch*.

Mappings & Inferences: I mapped contractual candidates using domain terms without relying on billed prices or labels. 30 selected invoices contain mapped inferences.

Confidence & Calibration: Confidence scores are assignable at 0.60 for fully verified calculations and 0.40 when inference exposure exists across patient records (assigned to 106 rows).

AI Assistance Transparency: An AI assistant substantially helped generate and refine code, unit tests, and prose based on my interactive prompts. Zero LLM calls are made at runtime.

Time Commitment & Future Enhancements

My hands-on execution time during final packaging was **6 hours and 30 minutes**. Given an additional week, I would focus on: (1) validating inferred mappings and unresolved units, (2) implementing comparable H1 pricing rules to calibrate row-level confidence, (3) formalizing boundary clause reviews, and (4) expanding rule coverage across remaining hospital scope.